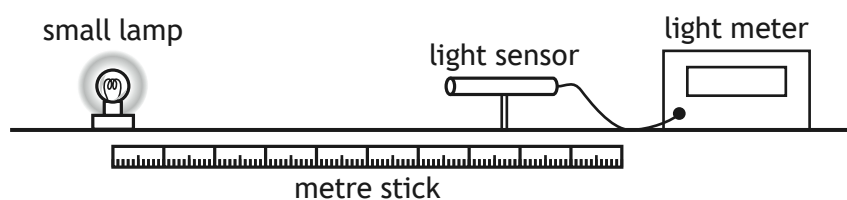


7. A student sets up an experiment, in a darkened room, to investigate the relationship between irradiance I and distance d from a point source of light. The apparatus used is shown.



The distance between the lamp and the light sensor is measured with a metre stick. The irradiance is measured with a light meter.

The results obtained are shown in the table.

d (m)	0.200	0.400	0.600	0.800
I (W m^{-2})	198.4	49.6	22.0	12.4

- (a) State what is meant by the term *irradiance*.

1

- (b) Use **all** the data to establish the relationship between irradiance I and distance d .

3

Space for working and answer

7. (continued)

- (c) Other than repeating each measurement and calculating the mean, suggest **two** improvements that could be made to the student's experimental procedure.

2

- (d) A second student carries out the experiment using a laser instead of a lamp.

- (i) The laser beam is initially shone onto the light sensor from a distance of 0.20 m.

The beam produces a uniformly lit circle of diameter 1.00×10^{-3} m on the light sensor.

The irradiance of the circle of light is $1.51 \times 10^3 \text{ W m}^{-2}$.

Determine the power of the laser beam.

4

Space for working and answer

[Turn over



* X 8 5 7 7 6 0 1 2 5 *

7. (d) (continued)

- (ii) The student moves the laser to a distance of 0.40 m from the sensor.
The reading on the light meter does not change.
Suggest why the reading does not change.

1

